HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Actuarial Mathematics and Statistics

F78PB

Probability and Statistics B

Semester 2 2021/22

Duration: Two Hours

ANSWER ALL QUESTIONS

Total Marks: 60

This is an open book exam but all answers must be in your own words to demonstrate your understanding. You must also show your working. Answers that do not demonstrate understanding or show working may not receive marks.

Upload time is for the purpose of scanning and uploading your answers. You must not use that time to continue working on your answers.

1. Suppose that X is a random variable with density $f(x) = e^{1-x}$ for $x \geq 1$ (density $f(x) = 0$ otherwise).

(a) Use the definition of $M_X(\theta)$, the moment generating function of X, to show that

$$M_X(\theta) = \frac{e^{\theta}}{1-\theta} \quad \text{for} \quad \theta < 1.$$

(4 marks)

(b) Use the moment generating function $M_X(\theta)$ given above to determine E(X).

(2 marks)

(c) Suppose that $Y = 2X - 2$. Show that

$$M_Y(\theta) = e^{-2\theta} M_X(2\theta)$$

where $M_X(\theta)$ and $M_Y(\theta)$ are the moment generating functions of X and Y respectively. Hence, or otherwise, show that $Y \sim Exp(\frac{1}{2})$. Be sure to give reasons.

(4 marks)

[Total 10 marks]

2. Consider a random sample of size n, $\underline{X} = (X_1, X_2, \ldots\ldots, X_n)$ from a distribution having probability density function

$$f(x;\theta) = e^{\theta - x} \text{ for } \theta \leq x < \infty$$

(a) Find the method-of-moments estimator (MME) of θ.

Hint: $\lim_{x\to\infty} xe^{\theta-x} = 0$

(5 marks)

(b) Give an equation for both the likelihood function $L(\theta;\underline{x}) = \prod_{i=1}^{n} f(x_i, \theta)$ and the log-likelihood function $l(\theta;\underline{x})$. By justifying that the log-likelihood function is an increasing function of θ, or otherwise, show that the maximum likelihood estimator (MLE) of θ is

$$\hat{\theta} = \max\{X_1, X_2, \ldots X_n\}$$

(4 marks)
[Total 9 marks]

3. Individuals in a group of 7 asthma sufferers were each given a new drug to try and relieve their condition. The time until each patient's next attack, in weeks, is thought to be modelled by a Gamma(2, β) distribution where β is unknown. All 7 sufferers were given the drug on the same day and each person is monitored over the following 8 weeks. In this time 5 patients have an attack at times $x_1,\ldots,x_5$ for which $\sum x_i = 22.6$. The remaining 2 patients do not have an attack.

(a) Show that the probability that a given patient does not have an attack during the monitoring period is given by $p(\beta) = (1 + 8\beta)e^{-8\beta}$.

(4 marks)

(b) Explain carefully why the likelihood satisfies $L(\beta) \propto \beta^{10}(1 + 8\beta)^2 e^{-38.6\beta}$.

(4 marks)

(c) Find a quadratic equation satisfied by the MLE of β. You do NOT need to solve this equation.

(4 marks)

[Total 12 marks]

4. Consider two normally distributed populations with means μ_1 and μ_2 and variances σ_1^2 and σ_2^2 respectively. Also consider two independent random samples, one from each population, given by $X_{11}, X_{12}, \ldots\ldots, X_{1n_1}$ and $X_{21}, X_{22}, \ldots\ldots, X_{2n_2}$ respectively.

(a) Show that $(\overline{X}_1 - \overline{X}_2)$ is an unbiased estimator for $(\mu_1 - \mu_2)$. (2 marks)

(b) Explain why, when the variances are known, the function

$$h(\underline{X}_1, \underline{X}_2, \mu_1 - \mu_2) = \frac{(\overline{X}_1 - \overline{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

is a pivotal quantity for the difference between $(\mu_1 - \mu_2)$.

(Note: You may quote any standard results) (4 marks)

(c) Suppose now that $\bar{x}_1 = 65$, $\bar{x}_2 = 70$, $n_1 = 12$, $n_2 = 10$ but this time the population variances are NOT known. We do know that $s_1 = 4.5$ and $s_2 = 3.1$. Calculate an equal-tailed 95% confidence interval for the difference between the population means specifying the pivotal quantity used in this case. (4 marks)

[Total 10 marks]

5.

One way to measure a person's fitness is to measure their body fat percentage. Data are measured for a random selection of people who did two different workout plans at a gym three times a week for a three-month period; at the end of which their trainer measured the body fat. The table below shows the data.

PLAN A

8.5	20.8	23.1	22.6	24.1	15.5	15.4	13.7

PLAN B

24.6	6.8	10.4	10.3	16.4	8.3	13.2	17.7

(a) Perform a hypothesis test to check whether or not there is a difference between the mean body fat percentage for Plan A and the mean body fat percentage for Plan B. Make sure you state the p-value and give a clear conclusion. You are told that $s_p^2 = 32.51$.

(5 marks)

(b) Perform a hypothesis test to check whether or not the population variances are equal. You are told that $s_A^2 / s_B^2 = 0.89$.

(3 marks)

(b) What are the main assumptions in assessing whether or not the test of part (a) is valid? Give reasons whether or not these assumptions hold for the above data.

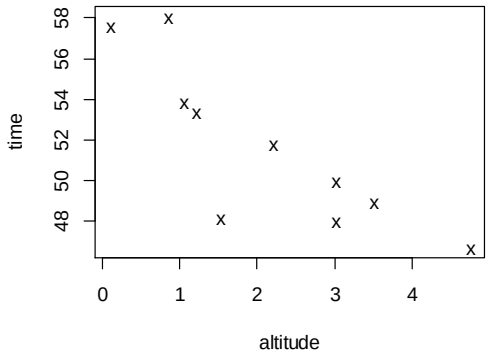
(2 marks)

[Total 10 marks]

6. The data below shows the record-winning times (minutes) for similar athletes running in 10 similar hill races in various parts of the world, against the altitude (km) of the region in which the race took place. The races all cover the same distance. Also shown is a scatter diagram of the results.

time (y)	50.0	53.4	51.8	48.2	49.0	48.0	46.7	58.1	53.9	57.6
alt (x)	3.00	1.20	2.20	1.52	3.50	3.00	4.75	0.85	1.05	0.10

Note $\sum x = 21.17, \quad \sum x^2 = 63.238, \quad \sum y = 516.7, \ \sum y^2 = 26842.51 \ \sum xy = 1050.369$



(a) Calculate the sample correlation coefficient for the data and, by making reference to the graph, comment briefly on any association between the record time for a hill race and the altitude at which it takes place. (3 marks)

(b) Test $H_0 : \rho = -0.8$ vs $H_1 : \rho < -0.8$. (4 marks)

(c) Without performing any further arithmetic calculations for this example show that, in general, the fitted least-squares regression line $\hat{y} = \hat{\alpha} + \hat{\beta}x$ passes through the point $(\bar{x}, \bar{y})$. (2 marks)

[Total 9 marks]